

VIVA CHEAT SHEET

Impact of UPI-Based Digital Payments on the Velocity of Money and Economic Activity in India

Everything you need to defend the project, in the order you are likely to need it

ROUNAK ARCHANA KAMESWARAN · Class XII · Economics

How to use this. Learn Parts 1 and 2 until you can say them without looking — between them they cover most of what an examiner actually asks. Part 3 is the general question bank. Part 4 is the five hard questions and deserves the most preparation, because those are the ones an examiner asks when a project is good. Part 5 covers opinion and risk questions, and Part 6 is what to do when you are stuck. Page numbers given as "p.24" refer to the printed pages of the research paper.

Part 1 · The opening answer

Almost every viva starts with some version of "tell me about your project". Have this ready. It is about sixty seconds spoken, and it sets the agenda — examiners usually ask their next question about something you just said, so every sentence here is a door you are choosing to open.

Say this: My project asks whether UPI has changed the speed at which money circulates in India. UPI grew from about 2 crore transactions in FY 2016-17 to over 24,000 crore in FY 2025-26. But the interesting part is not the size — it is that the average payment got smaller, from about ₹1,838 to ₹1,300. That tells me UPI did not just digitise payments that already existed; it added new, very small everyday payments. Using Fisher's equation of exchange, more transactions on the same money stock means a higher velocity of money. I also found that the value settled over UPI rose from about half of India's GDP to about nine-tenths of it in three years. On economic activity my conclusion is more cautious: 57 per cent of small merchants reported higher sales after adopting digital payment, which is suggestive, but I cannot prove UPI caused it because smartphones, internet and incomes all rose in the same decade.

Notice what that answer does. It names a concept (Fisher's identity), gives three specific numbers, states a finding that is genuinely your own, and admits a limit. Examiners reward all four.

The single sentence, if you only get one

Say this: UPI made very small payments worth making, and because there are now far more payments on roughly the same stock of money, money circulates faster.

Part 2 · Numbers to know cold

You will not be asked all of these. You will be asked some of them, and hesitating over a number in your own project looks worse than not knowing a concept. Learn the first block properly; skim the rest.

The essential eight

What	Number	Page
UPI volume, FY 2016-17 → FY 2025-26	2 crore → 24,162 crore	24
UPI value, FY 2016-17 → FY 2025-26	₹0.07 → ₹314 lakh crore	24
Average payment, peak → latest	₹1,838 (FY 2020-21) → ₹1,300	26
UPI value ÷ nominal GDP	0.52× → 0.91× in three years	30
UPI share of retail digital volume	81% (FY 2024-25)	28
India's share of world real-time payments	about 49% (2024)	36
Small merchants reporting higher sales	57%	19
Rural vs urban internet per 100 people	46.73 vs 113.83	38

The supporting set

What	Number
Banks live on UPI	44 (FY 2016-17) → 703 (FY 2025-26)
UPI users onboarded	55.49 crore (June 2026)
Merchants accepting UPI	about 6.5 crore; 56.86 crore QR codes
Composition of UPI volume	P2M is 63%; 86% of P2M is below ₹500
RBI Financial Inclusion Index	64.2 (Mar 2024) → 67.0 (Mar 2025)
RBI Digital Payments Index	465.33 → 493.22 (March 2018 = 100)
PMJDY beneficiaries	55.98 crore; over 55% women
Demonetisation	₹15.4 trillion withdrawn = 86.9% of note value; 98.96% returned
Digital payments around demonetisation	71.27 → 111.45 crore a month (+56%)

What	Number
UPI growth, pandemic years	78% (FY 2020-21), then 106% (FY 2021-22)
Total digital payments	2,071 crore (FY 2017-18) → 18,737 crore (FY 2023-24), CAGR 44%
Cybersecurity incidents	10.29 lakh (2022) → 22.68 lakh (2024)
Users reporting they spend more	about 75% in one survey
MDR on small-merchant UPI	zero up to ₹2,000; Government pays 0.15% incentive

The eight data sets in one line each

If an examiner points at any exhibit and says "explain this", these are your answers. They are the same sentences printed in the call-out boxes in the paper.

Data set	What it shows
1 · p.24	UPI grew about 12,000-fold in volume, and volume grew faster than value every year from FY 2021-22 — so payments were getting smaller.
2 · p.26	The average payment fell from ₹1,838 to ₹1,300 while volume rose ten-fold. This is my most important finding.
3 · p.28	UPI is 81% of retail digital transactions, so retail payment trends in this period can fairly be attributed to it.
4 · p.30	UPI value went from 0.52 to 0.91 times GDP. The level means little; the rise means a lot.
5 · p.32	Demonetisation gave a 56% jump, but 99% of the cash came back — it accelerated adoption without causing it.
6 · p.34	COVID growth was 78% then 106% — faster after the shock than during it, so the habit stuck.
7 · p.36	Financial Inclusion Index 64.2 → 67.0, with usage rising fastest, which is the sub-index that matters.
8 · p.38	Rural internet is 46.73 per 100 against urban 113.83 — the limit is connectivity, not acceptance.

Part 3 · Question bank

A. Topic and motivation

Q. Why did you choose this topic?

Because I use UPI every day and wanted to know whether something that changed my own habits had changed the economy. The velocity of money is a concept from the syllabus that nobody can observe directly, and UPI gave me a way to look at it with real published data.

Q. What exactly is your research question?

Two questions. First, has UPI changed the speed at which money circulates? Second, can any effect on real economic activity be identified from published data? I answer the first more confidently than the second, and the paper says so.

Q. What is new in your project? Everyone knows UPI grew.

Agreed — the growth is not the finding. My contribution is the average transaction size in Table 2. That figure is not published anywhere; I derived it by dividing annual value by annual volume. It falls from ₹1,838 to ₹1,300, and that decline is what distinguishes "UPI added new payments" from "UPI digitised old ones".

Q. What period does your study cover, and why?

FY 2016-17 to FY 2025-26. FY 2016-17 is UPI's first full year, and FY 2025-26 was the most recent complete year when I wrote. Ten full years also lets me look at two shocks — demonetisation and COVID-19.

B. Concepts

Q. What is the velocity of money?

The average number of times one unit of currency is used to buy goods and services in a given period. It is a behavioural property, not a physical one — it describes how quickly people part with money they hold.

Q. State the equation of exchange and explain each term.

$M \times V = P \times T$. M is the money stock, V is transactions velocity, P is the average price level, T is the volume of transactions. It is an identity — true by construction, not an assumption. Its use is that if M is roughly fixed and T rises, V must have risen.

Q. What is the Cambridge approach, and how does it relate to velocity?

It writes $M = k \times P \times Y$, where k is the fraction of income people hold as money. Velocity is the reciprocal of k , so $V = 1 \div k$. A rise in velocity and a fall in k are the same statement. This is the version I actually use, because UPI works by reducing how much cash people need to keep on hand.

Q. What is UPI, in one sentence?

A common standard, launched in 2016, that lets any bank account be paid from any app through a single virtual payment address — instantly, at any hour, at no cost to the user.

Q. What is NPCI, and why does it matter that it is not-for-profit?

The National Payments Corporation of India built and runs UPI under RBI guidance. Because it is not obliged to maximise a return on the switch itself, transactions can be routed at or near cost — which is why UPI is free and card payments are not.

Q. What is interoperability and why was it decisive?

It means a user of one app can pay a user of any other. Before UPI, wallets were closed loops — money in one could not reach another. The IMF studied UPI and found interoperability drove adoption, because users could join through a brand they trusted and later switch, which forced every provider to improve.

Q. Why did cards never achieve what UPI did?

Cost of acceptance. A card terminal costs money and the merchant pays a discount rate on every sale, which does not work on a ₹20 margin. A printed QR code costs nothing and, below ₹2,000, carries no fee at all.

Q. What is the difference between nominal and real GDP, and which did you use?

Nominal GDP is measured at current prices, so it includes inflation; real GDP is at constant prices. I used nominal GDP in Table 4, because UPI transaction value is also in current rupees — both sides of the ratio must be measured the same way.

C. Method

Q. What kind of data did you use?

Entirely secondary data — published statistics from RBI, NPCI, PIB, MoSPI and TRAI, plus research from the IMF, BIS and World Bank.

Q. Why did you not conduct a primary survey?

Because the thing I am studying is national in scale and is already measured comprehensively by the institutions that run it. A survey I could realistically

conduct would be too small to be representative, and could not improve on RBI and NPCI data.

Q. How did you make sure your sources were reliable?

I took every number from the institution that produced it rather than from news or commercial websites, and each figure carries a bracketed reference to the Bibliography so it can be checked.

Q. Which figures are your own calculations?

Two. The average value per transaction in Table 2, and UPI turnover as a multiple of GDP in Table 4. Both derivations are stated under their tables so anyone can reproduce them.

Q. What are the limitations of your study?

Seven are listed on page 14. The three that matter most: it is all secondary data; correlation is not causation, since smartphones and incomes rose alongside UPI; and velocity cannot be measured directly, so Table 4 is a proxy rather than a measurement.

Q. What were your hypotheses?

Three. H1 — transactions velocity has risen. H2 — UPI added genuinely new small payments rather than only migrating old ones. H3 — the gains are unevenly distributed between rural and urban India. All three are supported, H1 with a qualification.

D. Findings

Q. What is your most important finding?

That the average UPI payment shrank from ₹1,838 to ₹1,300 while volume rose more than ten-fold. If UPI had only digitised existing payments, average size would have stayed broadly flat. It fell — so UPI reached transactions it did not previously serve: the tea, the vegetables, the auto fare.

Q. Why does a falling average transaction size matter for velocity?

Because Fisher's identity is about the number of transactions, not their size. Adding a very large number of very small payments raises T sharply while adding little to total value — which is precisely an increase in how often each rupee is used.

Q. Did demonetisation cause UPI's growth?

It accelerated it but did not cause it. Digital payments rose 56 per cent in the seven months after November 2016, but about 98.96 per cent of the demonetised notes came back and cash use recovered. Growth continued for

years afterwards at rates far above 2016-17. Demonetisation removed the effort of trying something new; it did not change the economics.

Q. Why did UPI grow during COVID-19 when the economy shrank?

Two forces. Substitution — payments that would have been cash became digital. And adoption — merchants and customers who had resisted learned because there was no alternative. Adoption dominated, and I can show that: growth was 78 per cent in FY 2020-21 but 106 per cent in FY 2021-22. If it had been pure substitution it would have reversed once shops reopened. It did not.

Q. Why is demonetisation's effect temporary but COVID's permanent?

Duration. Demonetisation removed the alternative for a few months, so people complied and then reverted. The pandemic lasted long enough for digital payment to become a habit, and habits survive the removal of the constraint that formed them.

Q. Has UPI increased financial inclusion?

Yes, measurably. The RBI Financial Inclusion Index rose from 64.2 to 67.0 in one year, and importantly the usage sub-index rose fastest. Access was already largely solved by Jan Dhan — 55.98 crore accounts. The problem was dormancy, and the World Bank records the share of Indian women with inactive accounts falling from a third in 2021 to 18 per cent in 2024.

Q. Has UPI increased GDP?

I cannot claim that, and I deliberately do not. What I can say is that 57 per cent of small merchants reported higher sales after adopting digital payment, and that the BIS finds fast payment systems raise digital finance adoption most in lower-income economies. That is evidence pointing towards a real effect, not a measurement of one.

Part 4 · The five hard questions

These are the questions an examiner asks when the project is good enough to probe. Each targets a place where the paper is more careful than it strictly needed to be — which is exactly why they are worth marks if you handle them well. Prepare these five above all else.

Q. Hard 1. Which velocity have you measured — income or transactions?

Transactions velocity, and the difference matters. Income velocity counts only payments for newly produced output, so it moves slowly. Transactions velocity counts every payment, including transfers between my own accounts and payments between wholesalers. A payment system acts on transactions velocity directly and on income velocity only indirectly. Most popular writing confuses the two, so I labelled mine explicitly.

Q. Hard 2. So does 0.91× mean every rupee circulates 0.91 times a year?

No, and that is the trap in the number. The top of the ratio counts every payment; the bottom counts only final output. They are not commensurable, so the level of the ratio is not meaningful. What is meaningful is the rate of change: the mismatch is broadly constant year to year, so a rise from 0.52 to 0.91 in three years tells me payment activity grew much faster than output.

Q. Hard 3. Some of the rise is just cash payments becoming visible. Doesn't that ruin your finding?

It qualifies it, and I say so in the paper. A cash payment that becomes a UPI payment raises measured turnover without any change in behaviour, because cash was never counted. The available data cannot separate that from genuinely new transactions. But Data Set 2 gives me an independent check: if this were only migration, average payment size would have held steady. It fell by 29 per cent, so at least part of the rise is genuinely new activity.

Q. Hard 4. Inflation was positive over your period. Doesn't that undermine the falling average?

It strengthens it. Rising prices push the average value of any payment up, so inflation works against my finding rather than producing it. The average fell anyway — which means the shift towards small payments is larger in real terms than the nominal figures show.

Q. Hard 5. If you cannot prove causation, what have you actually established?

Three things. That transactions turnover per rupee of output rose sharply — that is measured, not inferred. That the growth consists of new small payments, not just migrated ones — that is the average-size evidence. And that the effect on output is supported by merchant and cross-country evidence but not proved, because smartphones, internet access, bank accounts and incomes all expanded in the same decade and published aggregates cannot separate them. Refusing to overclaim is a finding, not a gap.

If pressed further on Hard 5: To prove causation I would need district-level UPI data and a source of variation — for example comparing districts where adoption was forced early against otherwise similar ones. The IMF did something like this and found faster growth where interoperability helped most. That data is not published, which is why one of my six suggestions is that disaggregated payment statistics should be released.

Part 5 · Opinion, risks and the awkward ones

Q. Is UPI being free a good thing? Someone must be paying.

The exchequer pays, through incentive schemes worth about ₹1,500 crore for low-value merchant transactions. Whether that is worth it depends on whether the gains in formalisation, inclusion and tax visibility exceed the outlay — which is why one of my suggestions is that the scheme should be evaluated publicly and periodically.

Q. What are the risks of a cashless economy?

Three. Infrastructure dependence — a digital payment needs power, a device and a network at the same time, and cash needs none of them, so an outage stops commerce. Fraud — cyber incidents doubled from 10.29 to 22.68 lakh between 2022 and 2024. And exclusion — a system reachable only through the internet cannot be more inclusive than internet access itself.

Q. Is UPI secure?

The technology is strong — device binding, a compulsory PIN, and machine-learning fraud monitoring that declines suspicious transactions. The weak point is the user, who can be talked into authorising a payment. That is why my suggestion is fraud awareness delivered inside the apps in regional languages, rather than more technical security.

Q. Does UPI make people overspend?

The evidence suggests it can. In one survey about 75 per cent of users reported spending more, many saying digital money felt less tangible than cash. It is a real cost, especially for low-income households, and I treat it as one — the same frictionlessness that raises velocity removes a restraint cash imposed without anyone designing it.

Q. Why is rural adoption behind? Is it a lack of QR codes?

No — that is the point of Data Set 8. Over 5.45 crore acceptance touch points have been deployed in tier-3 to tier-6 centres. The constraint is connectivity: rural internet is 46.73 subscribers per 100 people against an urban 113.83. So extending broadband is a more effective payments policy at the margin than more acceptance subsidy.

Q. India has 49 per cent of the world's real-time payments. Isn't that just population?

Partly, and I say so. Volume share is not a measure of quality. What makes it notable is the policy difference: most fast payment systems abroad charge merchants, and India chose not to. The adoption is partly a consequence of treating payment as a public utility.

Q. What would you do differently or study next?

I would want district-level data to examine regional variation, which national aggregates hide completely. If I extended the project, I would look at whether the payment record itself is changing credit access for small traders through the Unified Lending Interface — that is where a payment effect would turn into an investment effect.

Q. Did you write this yourself?

Yes. Answer plainly, then show it: name the two calculations you derived, say why you chose nominal rather than real GDP, and explain the base-year break in Table 4. Nobody who has not engaged with the material can do that.

Part 6 · Traps, and how to get unstuck

Six things not to say

Do not say	Say instead
"UPI increased India's GDP."	"The evidence points that way — 57% of small merchants reported higher sales — but I cannot prove causation."
"Each rupee now circulates 0.91 times."	"The level of that ratio is not meaningful. Its rise from 0.52 to 0.91 is."
"Velocity increased" (unqualified)	"Transactions velocity increased. Income velocity would move far less."
"Demonetisation made India digital."	"It accelerated adoption, but 99% of the cash returned. The lasting change came later."
"Cash is finished."	"Cash is declining as a share of payments, but currency in circulation is still large relative to GDP."
"My data is from the internet."	"My data is from RBI, NPCI, PIB, MoSPI and TRAI publications, each referenced in the Bibliography."

If you do not know the answer

You will be asked something you have not prepared. That is normal and is not a failure. What examiners penalise is bluffing, because it is obvious. Three usable moves:

1. Say what you do know, then mark the limit honestly. "I did not examine that directly. What my data does show is..." — this turns an unknown into an answer.
2. Reason out loud from the identity. Almost every velocity question can be worked from $M \times V = P \times T$. Say what happens to each term and why. Visible reasoning scores better than a remembered fact.
3. Point to the paper. "That is in my limitations on page 14" or "Data Set 8 covers that" is a legitimate answer, and it shows you know your own structure.

The one habit that matters: Whenever you state a number, say where it came from — RBI, NPCI, PIB, TRAI, the IMF. It takes three extra words, it

demonstrates the project is genuinely researched, and it makes every answer harder to challenge.

The last thing to read before you go in

Remember: UPI grew from 2 crore to 24,162 crore transactions. The average payment fell from ₹1,838 to ₹1,300. Value settled rose from 0.52 to 0.91 times GDP. Small payments got added, so money moves faster. On output, the evidence is suggestive and I do not overclaim. Everything else is detail.